Bayesian Marketing Mix Prototype

A reproducible foundation for forecasting, channel attribution, budget allocation, and learning.

Built and tested: 156 weeks x 4 products x 3 price tiers (1,872 observations); 130 training weeks and 26 held-out weeks. Baseline, stockout-only, funnel-only, and combined variants passed predefined gates; 12 deterministic regression tests passed.

Workflow: generate known truth > fit MCMC > validate forecasts/effects > attribute sales > optimize > export reports

MODEL: WHAT IT REPRESENTS

Demand and hierarchy. Latent log sales combine global, product, tier, and product x tier intercepts with partial pooling. Controls cover price, macro demand, base-price growth, promotion, post-promotion dip, PrimeDays, and annual seasonality. No random-walk baseline.

Media response. Display, Video, Social, and CTV each have estimated geometric carryover (current week + 12 lags), saturation, and nonnegative log-sales coefficients. Social has an S-shaped Hill curve; the others have diminishing-return Hill curves. Paid clicks have saturation but no adstock. Response coefficients are shared across products/tiers.

Funnel mediation. CTV plus observed organic demand and controls drive latent branded search. Search has log-Normal measurement noise; clicks are Binomial conditional on searches, with spend-dependent capture. Sales use latent mean clicks, not held-out mediator outcomes. CTV also has a direct sales effect.

Inventory and sales. Observed sales = min(demand, supplied inventory). A censored log-Normal likelihood learns demand behind capped sales. Forecasts distinguish latent demand from fulfillment; scenario evaluation uses E[min(demand, cap)], not a clipped mean.

Attribution and decisions. An explicit accounting rule assigns CTV direct plus mediated log lift and paid search its no-CTV click lift. Contributions reconcile to sales without double counting. A 13-week optimizer maximizes expected fulfilled units at fixed budget, constant weekly shares, fixed product proportions, and channel bounds, retaining carryover.

COMBINED RUN: EVIDENCE

Sampling reliability
R-hat 1.0045; 0 divergences

Holdout log-sales error
87.6% lower RMSE*

90% predictive coverage
Demand 89.1%; sales 92.9%

90% mediator coverage
Search 91.3%; clicks 94.2%

Known media-effect recovery
5/5 intervals cover truth; max error 0.83%

CTV mediated lift recovery
2.7% relative error

13-week allocation scenario
$5.72m budget; +0.49% units

*Versus a product-tier mean log-sales benchmark. All gates passed. Budget conserved; allocation objective shortfall is 0.0001% versus the known-truth best found under the same constraints. Six local starts are not a global-optimum proof.

NEXT STEPS: MMM BEST PRACTICE AND QUANT RESEARCH

1. Identification and experiments. Calibrate with geo/lift studies; address spend endogeneity, mediator confounding, and measurement error before causal interpretation.

2. Robustness and backtesting. Repeat seeds and leakage-free rolling/untouched holdouts; add simulation-based calibration, prior sensitivity, and misspecification stress tests.

3. Real-data pilots. Audit exposure, spend, sales, and lag alignment; use stronger forecast benchmarks and explicit future-control, organic-demand, and inventory scenarios.

4. Quant decision layer. Optimize margin/profit over posterior scenarios with tail-loss (CVaR) risk constraints; research dynamic pacing, uncertain CPM/CPC, and replenishment.

5. Dynamic response research. Test time-varying/hierarchical effects, alternative lag and saturation models, and state-space baselines; justify complexity by decision value.

Boundary: one deliberately informative synthetic seed with matching model assumptions. Reference variants preserve comparability; the paired holdout is not independent new evidence. Future drivers/inventory are supplied, not forecast. Attribution is a convention; the objective is units, not profit. This is **not real-world causal validation or a deployment recommendation.**